

First Half of 2026 Freshmen Seminar (DFRC #51)

Freshmen Mock Case Scenario

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- ☐ **Introduction to Assignment**
- ☐ **My Answer Sheets**
- ☐ **Problem Analysis and Solution**
- ☐ **Conclusion**

Introduction to Assignment

Case Overview

■ Objectives

- Evaluation of basic techniques for digital data collection and analysis across various platforms
- Inferring case scenarios and reconstructing timelines based on the provided digital evidence

■ Evaluation Factors / Scoring

■ Evaluation Factors

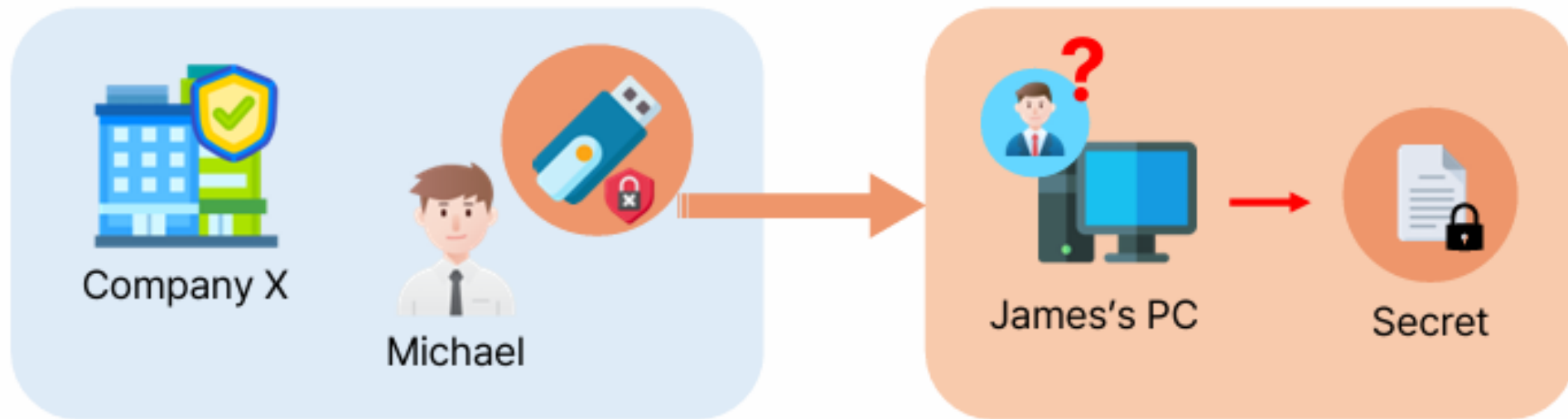
- Is the use of tools appropriate, and is the analysis process systematic and logical?
- Has the scenario been logically reconstructed based on the digital evidence?
- Was the report written clearly, and was the presentation easy to understand?

■ Scoring

- Problem Solving (60 pts) + Drafting Timeline & Visual Summary (20 pts) + Presentation (20 pts)

Case Overview

- **Mission: Reveal the truth behind the defense company X's technology leakage case!**
 - Michael, the employee of X, has been captured from carrying out unauthorized USB
 - During investigation, important files that only James can read were found in Michael's PC
 - Now, James is also suspected of being a suspect in the technology leakage case

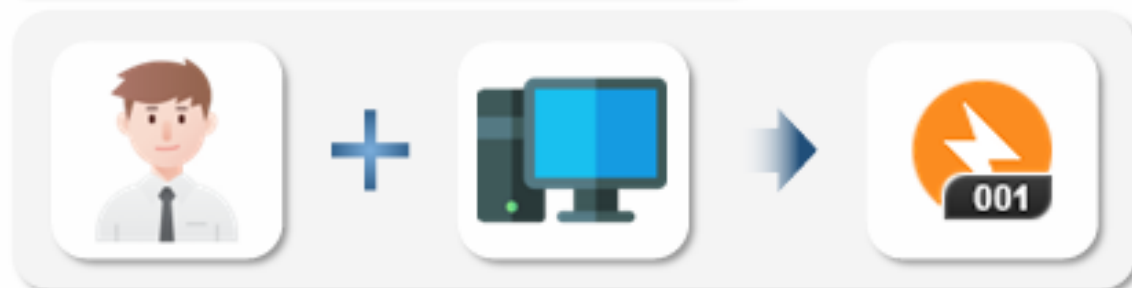


- You have to figure out the case's timeline, and reveal the truth behind of it!

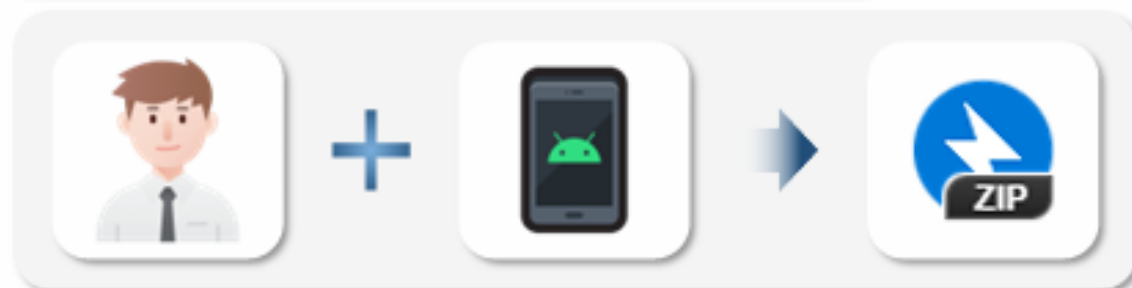
Case Overview

- Investigation Materials

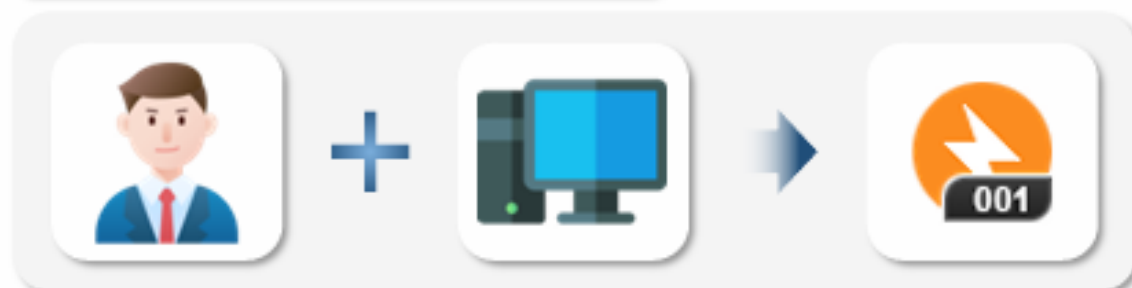
① Michael's PC (Michael_PC.001)



② Michael's Phone (Michael_Mobile.zip)



③ James' PC (James_PC.001)



④ AI Server PC (Server_PC.001)



My Answer Sheets

My Answer Sheets

No.	Answer	No.	Answer
Q1	① Daniel's business card information - Name: Daniel - Company & Role: Director of Strategic Programs Defense System Division - Email: daniel.northridgedefense@gmail.com - Phone Number: 010-6281-9220 ② James' PC Password: rltnfrogkf1!	Q4	STEP 1. Michael connects RDP from his PC to James PC STEP 2. AI Server IP & Password found on James PC STEP 3. RDP connection from James PC to AI server STEP 4. Get conversation logs and new product or technology related records from LM Studio from the AI server to Drag & Drop to Michael PC STEP 5. Exporting from Michael PC to the outside via USB
Q2	① Gmail, Telephony, MMS & SMS ② ChangeNow ③ 0x62ff5519fb2d8 ④ 0xf2e5f287a02d0b10cb672bf6a99f237bb01959c4 ⑤ PEPE, ETH (Get PEPE deposit and switch ETH from ChangeNow) ⑥ Transaction creation time: 2026-01-14T16:02:33 PEPE received time: 2026-01-14T16:16:32 ETH changed time: 2026-01-14T16:17:42 ⑦ Daniel's Criminal Plan - Scheduled Date: 2026.01.15 - Target: Product development-related information - File Types: Any available format - Transfer Method: USB - Note 1. Execute the operation after all employees have left the office 2. Leave no unnecessary traces	Q5	(Short-answer question and fill in the table in the slide 7) <i>- described later</i>
Q3	Information about an AI server 1. AI Platform used: LM Studio (from Web History, Web Download) 2. Found artifacts - Downloaded models: Qwen 2.5-based custom models (from C:/Users/Administrator/.lmstudio/.internal/download-jobs-info.json) - Model Information: (from C:/Users/Administrator/models/) - Conversation history: We can view conversation history that asks LLM to write a report based on minutes James have written (from C:/Users/Administrator/.lmstudio/conversations/) - Upload file: ITCS-X Meeting_Minutes.docx (from C:/Users/Administrator/.lmstudio/user-file/)	Q6	(Fill out the given template) <i>- described later</i>

Problem Analysis and Solutions

Questions

- Q1. Based on the MMS conversation between James and Michael, identify and describe the following information obtained from the exchanged messages.

- Requirements

- Daniel's business card information
- James's PC password

- Hint

- The password for James' PC shared through MMS is stored in png format on Michael's Mobile.

Solution to Question 1

■ STEP 1) Michael's Phone Analysis

- Michael's Phone was analyzed based on **ALEAPP**.
- The reason is that **adb** could be observed in the Dump file, and the directory structure was **Android's directory structure**.



adb

anr

apex

app

app_fonts

app-asec

2026-01-21 오후 7:36 파일 폴더

2026-01-21 오후 7:40 파일 폴더

2026-01-21 오후 7:40 파일 폴더

2026-01-21 오후 7:45 파일 폴더

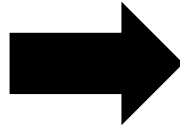
2026-01-13 오후 4:10 파일 폴더

2026-01-13 오후 4:10 파일 폴더

Solution to Question 1

■ STEP 2) Using ALEAPP

_HTML	2026-01-21 오후 10:57	파일 폴더	
_Timeline	2026-01-21 오후 10:57	파일 폴더	
_TSV Exports	2026-01-21 오후 10:57	파일 폴더	
data	2026-01-21 오후 10:54	파일 폴더	
Script Logs	2026-01-21 오후 10:54	파일 폴더	
_Java_artifacts.db	2026-01-21 오후 10:55	Data Base File	80KB
_Java_data.json	2026-01-21 오후 10:57	JSON 원본 파일	4KB
index.html	2026-01-21 오후 10:57	Chrome HTML Do...	1KB



ALEAPP 3.4.0

POWER EVENTS

- Last Boot Time
- Power Off Reset

RECENT ACTIVITY

- Recent Activity 0

SMS & MMS

- MMS messages**
- SMS messages

USAGE STATS

- UsageStats 0

WIFI PROFILES

- Wi-Fi Hotspot
- Wi-Fi Profiles
- Wifi Configuration Store Combine

WIPE & SETUP

- Appops.xml Setup Wizard
- Factory Reset

ALEAPP

MMS messages report

Total number of entries: 7

MMS messages located at: C:\Users\cgumg\Desktop\대학 4학년 2학기 방학\코딩 테스트\DFRC Seminar Coding Test\Week 3 Assignment\Week 3 Assignment\ALEAPP_Reports_2026-01-21_Wednesday_225448\data\Dump\data\user_de\0\com.android.providers.telephony\databases\mmssms.db

Show 15 entries

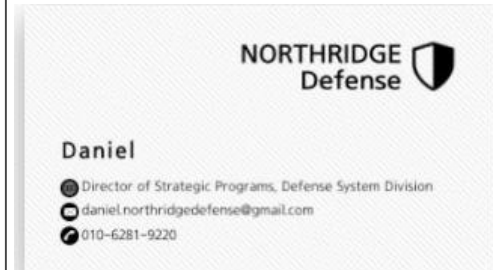
Search:

Date	MSG ID	Thread ID	Date sent	Read	From	To	Cc	Bcc	Body
2026-01-13 07:54:39	2	1	2026-01-13 07:54:38	1	01080186907	01055298452			Assistant Manager Michael, do you have any plans today? There's an important meeting, but I accidentally scheduled it at the same time as another meeting. Could you go in my place?
2026-01-13 07:56:18	4	1	2026-01-13 07:56:17	1	01080186907	01055298452			

NORTHEDGE Defense

Solution to Question 1

■ STEP 3) MMS, SMS analysis, and table creation

Sender	Receiver	Send Time	Receive Time	Message
01080186907	01055298452	2026-01-13 16:54:39	-	Assistant Manager Michael , do you have any plans today? There's an important meeting, but I accidentally scheduled it at the same time as another meeting. Could you go in my place?
01055298452	01080186907	2026-01-13 16:55:10	-	Yes, understood.
01080186907	01055298452	2026-01-13 16:56:12	-	It's with a potential external partner. Just meet with them and see how it goes. Here's the business card of the person attending today's meeting.
01055298452	01080186907	2026-01-13 16:56:16		Understood. I'll report back after the meeting.
01080186907	01055298452	2026-01-13 16:56:17		

Solution to Question 1

- STEP 3) MMS, SMS analysis, and table creation

01062819220	01055298452	2026-01-13 21:03:30		Thank you for the meeting today .
01062819220	01055298452	2026-01-13 21:03:49		There's no rush to decide on the proposal I mentioned.
01062819220	01055298452	2026-01-13 21:03:54		so please think it over and get back to me. If the circumstances are right, we can definitely meet your conditions.
01055298452	01062819220	2026-01-13 21:04:54		Yes, understood.
01055298452	01062819220	2026-01-14 15:10:24		Regarding the proposal you made at our last meeting, could you provide some more specific details?

Solution to Question 1

■ STEP 3) MMS, SMS analysis, and table creation

01055298452	01062819220	2026-01-14 15:11:05		If possible, I'd like to make a decision quickly due to personal circumstances...
01062819220	01055298452	2026-01-14 15:12:06		This isn't something I can explain upfront. Once you've made your decision, please contact me again.
01055298452	01062819220	2026-01-14 15:22:46		I've thought it over from various angles, and the direction you've proposed seems to be the only realistic option available to me. I'd like to move forward.
01062819220	01055298452	2026-01-14 15:25:33		Understood. I'll provide you with the detailed information via email going forward.
01080186907	01055298452	2026-01-14 17:08:29		Assistant Manager Michael , I'm on an external business trip right now, but the company VPN keeps disconnecting so I can't access my PC.
01080186907	01055298452	2026-01-14 17:09:05		There's a document I urgently need to check. Could you go to my desk and access my PC?
01055298452	01080186907	2026-01-14 17:10:02		Yes, that's possible. If you provide me with the PC password . I will log in.
01080186907	01055298452	2026-01-14 17:11:15		You know the method we use at our company, right? I've embedded it in the photo, so check it and log in.

Solution to Question 1

- STEP 3) MMS, SMS analysis, and table creation

01080186907	01055298452	2026-01-14 17:11:15		
01055298452	01080186907	2026-01-14 17:11:55		Yes, confirmed.

Solution to Question 1

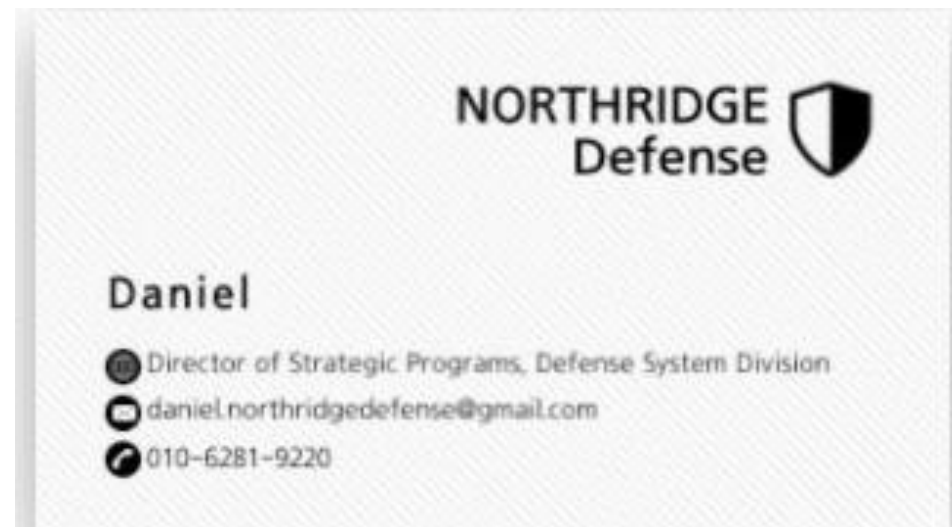
- STEP 3) MMS, SMS analysis, and table creation

01055298452	01080186907	2026-01-14 17:16:32		The document you mentioned does not exist on your PC.
01080186907	01055298452	2026-01-14 17:18:19		Oh, sorry. It should be on the USB on the desk. Please connect it and check it out.
01055298452	01080186907	2026-01-14 17:20:43		I have found it, I will send you the document.
01055298452	01080186907	2026-01-14 17:23:32		I sent it by gmail because it wasn't sent by text.
01080186907	01055298452	2026-01-14 17:27:24		I checked your mail. Thanks.

Solution to Question 1

■ STEP 3) MMS, SMS analysis, and table creation

- The following **facts** can be found when the information is put together.
 - 010-8018-6907: James
 - 010-5529-8452: Michael, **Assistant Manager**
 - 010-6281-9220: Daniel, **Director of Strategic Programs, Defense System Division, NORTHRIDGE Defense**



Solution to Question 1

■ STEP 4) Found James' PC Password

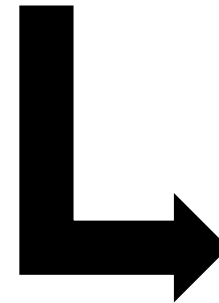
- We can find the file in '`DumpWdataWuser_deWOWcom.android.providers.telephonyWapp_parts`'
- However, the file has an **extension of jpeg**.

 PART_1768378275229_Resized_Mountain_5695156675535.jpeg

2026-01-21 오후 10:57

JPEG 파일

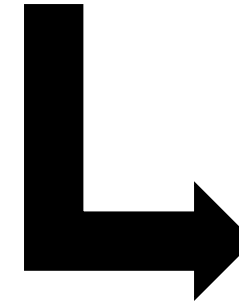
429KB



Solution to Question 1

- **STEP 4) Found James' PC Password**
 - We can find the file in '**DumpWdataWmediaWOWDownload**'

 Resized_Mountain_5695156675535.png	2026-01-15 오후 4:05	PNG 파일	3,905KB
 takeout-20260116T040536Z-3-001.zip	2026-01-16 오후 1:06	압축(ZIP) 파일	610KB



■ STEP 4) Found James' PC Password

-



Solution to Question 1

- STEP 4) Found James' PC Password
 - Further analysis revealed the presence of a **JPEG header**.
 - By **carving** and saving the header and footer, the extracted file was examined, which allowed the **password to be identified**.

```
003CC160 44 AE 42 60 82 FF D8 FF E0 00 10 4A 46 49 46 00 D0B`,y0yà..JFIF.
003CC170 01 01 00 00 01 00 01 00 00 FF E2 01 D8 49 43 43 .....yâ.ØICC
003CC180 5F 50 52 4F 46 49 4C 45 00 01 01 00 00 01 C8 00 _PROFILE.....È.
003CC190 00 00 00 04 30 00 00 6D 6E 74 72 52 47 42 20 58 ....0..mnrRGB X
003CC1A0 59 5A 20 07 E0 00 01 00 01 00 00 00 00 00 61 YZ .à.....a
003CC1B0 63 73 70 00 00 00 00 00 00 00 00 00 00 00 00 csp.....
003CC1C0 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 .....
003CC1D0 00 F6 D6 00 01 00 00 00 00 D3 2D 00 00 00 00 00 .öÖ.....ö-.....
.
.
.
003D0270 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
003D0280 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
003D0290 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
003D02A0 00 00 00 00 00 00 00 00 00 00 03 FF D9 .....yÜ
```

rltnfroqkf1!

Solution to Question 1

Q1	① Daniel's business card information - Name: Daniel - Company & Role: Director of Strategic Programs Defense System Division - Email: daniel.northridgedefense@gmail.com - Phone Number: 010-6281-9220 ② James' PC Password: rltnfroqkf1!
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Questions

- **Q2. Describe the criminal plan and cost negotiation between James and Michael based on the available digital evidence.**

- Requirements

- All communication methods used between Daniel and Michael
- Name of the cryptocurrency exchange service used
- Transaction ID
- Michael's wallet address
- Types of cryptocurrencies involved in the transaction
- Relevant date and time of the transaction or exchange attempt
- A detailed description of Daniel's criminal plan

Solution to Question 2

■ STEP 1) Gmail Analysis

- In the text messages, Michael accepts Daniel's proposal and suggests discussing the details via **Gmail**.
- Accordingly, the **Gmail was analyzed**.

01055298452	01062819220	2026-01-14 15:11:05		If possible, I'd like to make a decision quickly due to personal circumstances...
01062819220	01055298452	2026-01-14 15:12:06		This isn't something I can explain upfront. Once you've made your decision, please contact me again.
01055298452	01062819220	2026-01-14 15:22:46		I've thought it over from various angles, and the direction you've proposed seems to be the only realistic option available to me. I'd like to move forward.
01062819220	01055298452	2026-01-14 15:25:33		Understood. I'll provide you with the detailed information via email going forward.
01080186907	01055298452	2026-01-14 17:08:29		Assistant Manager Michael , I'm on an external business trip right now, but the company VPN keeps disconnecting so I can't access my PC.
01080186907	01055298452	2026-01-14 17:09:05		There's a document I urgently need to check. Could you go to my desk and access my PC?
01055298452	01080186907	2026-01-14 17:10:02		Yes, that's possible. If you provide me with the PC password , I will log in.
01080186907	01055298452	2026-01-14 17:11:15		You know the method we use at our company, right? I've embedded it in the photo, so check it and log in.

Solution to Question 2

STEP 1) Gmail Analysis

- The **takeout** file can be found in the directory '**Dump\data/media\WOW\Download**'
- Analysis of the compressed file revealed an '**.mbox file**', which was then examined using **MBox Viewer**.

takeout-20260116T040536Z-3-001.zip

↓

All mail Including Spam and Trash.mbox

↓

NN	보낸 사람	대상	받는 ...	참조	숨은 ...	제출 ...	배송 ...	생성 ...	수정 ...	첨부 파일
☐	0	Michael <michael.axisdef...	james.axisdefense@g...	2026-01-14 오후 5:24:18						
☐	1	Michael <michae...	Re:... Daniel <daniel.northr...	2026-01-14 오후 4:38:12						
☐	2									
☐	3	Daniel <daniel.n...	Re:... Michael <michael.axi...	2026-01-14 오후 4:35:45						
☐	4	Michael <michae...	Re:... Daniel <daniel.northr...	2026-01-14 오후 4:05:19						
☐	5									
☐	6	Daniel <daniel.n...	Re:... Michael <michael.axi...	2026-01-14 오후 3:55:55						
☐	7	Michael <michae...	Re:... Daniel <daniel.northr...	2026-01-14 오후 3:53:19						
☐	8									
☐	9	Daniel <daniel.n...	Plan Michael <michael.axi...	2026-01-14 오후 3:50:24						

Daniel
daniel.northridgedefense@gmail.com
2026-01-14 오후 3:50:24

Plan
끝: Michael <michael.axisdefense@gmail.com>

Let's proceed with the negotiation at this point. I am looking for information related to the new product. If you have a concrete plan for obtaining the data, please share it.

Solution to Question 2

- STEP 2) Gmail Table Creation and Analysis

Sender	Receiver	Send Time	Receive Time	Message
Daniel	Michael	2026-01-14 15:50:24	-	Let's proceed with the negotiation at this point. I am looking for information related to the new product. If you have a concrete plan for obtaining the data, please share it.
Michael	Daniel	2026-01-14 15:53:19		Regarding the information you mentioned, there is a high likelihood that internal AI servers contain materials generated during work processes, including conversation logs, as well as records related to new products or technologies. However, access to the server is required. Since my account does not have sufficient internal access privileges, I will need to use another PC to obtain the data. Could you please provide more details regarding the compensation?

Solution to Question 2

- STEP 2) Gmail Table Creation and Analysis

Daniel	Michael	2026-01-14 15:55:55		That sounds acceptable. Non-public internal materials related to the new product, including conversation records or files, will be sufficient. To avoid traceability, payment will be made using cryptocurrency. As discussed previously, an advance payment of 0.001% will be provided, and the remaining amount will be paid immediately upon successful delivery of the data. Please generate a one-time address through a web-based exchange and send it to me.
Michael	Daniel	2026-01-14 16:05:35		Understood. My wallet address is (Find it).
Daniel	Michael	2026-01-14		The payment has been completed. Please

Solution to Question 2

- STEP 2) Gmail Table Creation and Analysis

		16:35:45		confirm receipt , and proceed according to the plan while ensuring that no unnecessary traces are left . I have attached a file summarizing the plan once again. The file password is the numeric value of the last six characters of my wallet address.
Michael	Daniel	2026-01-14 16:38:12		Confirmed. I will contact you after the task is completed.
Michael	James	2026-01-14 17:24:18		ITCS-X Development Proposal.docx (42.13 KB)

Solution to Question 2

■ STEP 2) Gmail Table Creation and Analysis

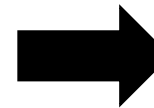
- From the email records, the following information can be identified:
 - Michael stated that he **does not have authorization** to access the AI server and therefore must **use another PC**.
 - The AI server is highly likely to contain **conversation logs or proprietary technical data**.
 - Daniel requires **technology-related materials**.
 - Michael and Daniel conducted their **transaction using cryptocurrency**.
 - Michael generated a **one-time password** through a **web-based exchange service**.
 - The password for the file containing Daniel's criminal plan consists of the **six numeric number** of his personal wallet.
 - James and Michael are employees of **Axis Defense**.

Solution to Question 2

■ STEP 3) Analysis of Michael PC

- **Analysis objective:** Since a web-based exchange site was used, access traces are expected to remain.
 - **Web Browser history** was analyzed.

1	1	https://www.google.com/search?...	changenow - Google 검색
2	2	http://changenow.io/	Instant Cryptocurrency Exchange ...
3	3	https://changenow.io/	Instant Cryptocurrency Exchange ...
4	4	https://changenow.io/?from=pepe&to=eth	Instant Cryptocurrency Exchange ...
5	5	https://changenow.io/?from=pepe&to=eth&amount=37800000	Instant Cryptocurrency Exchange ...
6	6	https://changenow.io/?from=pepe&to=eth&amount=3780000	Instant Cryptocurrency Exchange ...
7	7	https://changenow.io/?from=pepe&to=eth&amount=378000	Instant Cryptocurrency Exchange ...
8	8	https://changenow.io/?from=pepe&to=eth&amount=37800	Instant Cryptocurrency Exchange ...
9	9	https://changenow.io/?from=pepe&to=eth&amount=3780	Instant Cryptocurrency Exchange ...
10	10	https://changenow.io/?from=pepe&to=eth&amount=378	Instant Cryptocurrency Exchange ...
11	11	https://changenow.io/?from=pepe&to=eth&amount=37	Instant Cryptocurrency Exchange ...
12	12	https://changenow.io/?from=pepe&to=eth&amount=3	Instant Cryptocurrency Exchange ...
13	13	https://changenow.io/?from=pepe&to=eth&amount=	Instant Cryptocurrency Exchange ...
14	14	https://changenow.io/?from=pepe&to=eth&amount=8	Instant Cryptocurrency Exchange ...
15	15	https://changenow.io/?from=pepe&to=eth&amount=87	Instant Cryptocurrency Exchange ...
16	16	https://changenow.io/?from=pepe&to=eth&amount=872	Instant Cryptocurrency Exchange ...
17	17	https://changenow.io/?from=pepe&to=eth&amount=8725	Instant Cryptocurrency Exchange ...
18	18	https://changenow.io/?from=pepe&to=eth&amount=87255	Instant Cryptocurrency Exchange ...
19	19	https://changenow.io/?from=pepe&to=eth&amount=872557	Instant Cryptocurrency Exchange ...
20	20	https://changenow.io/exchange?from=pepe&to=eth&amount=872557	Crypto Swap with Lowest Fees & No ...
21	21	https://changenow.io/exchange/txs/62ff5519f9b2d8	Crypto Swap with Lowest Fees & No ...



2026-01-14 16:02:33 KST	https://changenow.io/exchange/txs/62ff5519f9b2d8
2026-01-14 16:02:33 KST	https://changenow.io/exchange/txs/62ff5519f9b2d8
2026-01-14 16:02:33 KST	https://changenow.io/exchange/txs/62ff5519f9b2d8
2026-01-14 16:00:58 KST	https://changenow.io/exchange?from=pepe&to=eth...
2026-01-14 16:00:58 KST	https://changenow.io/exchange?from=pepe&to=eth...

Solution to Question 2

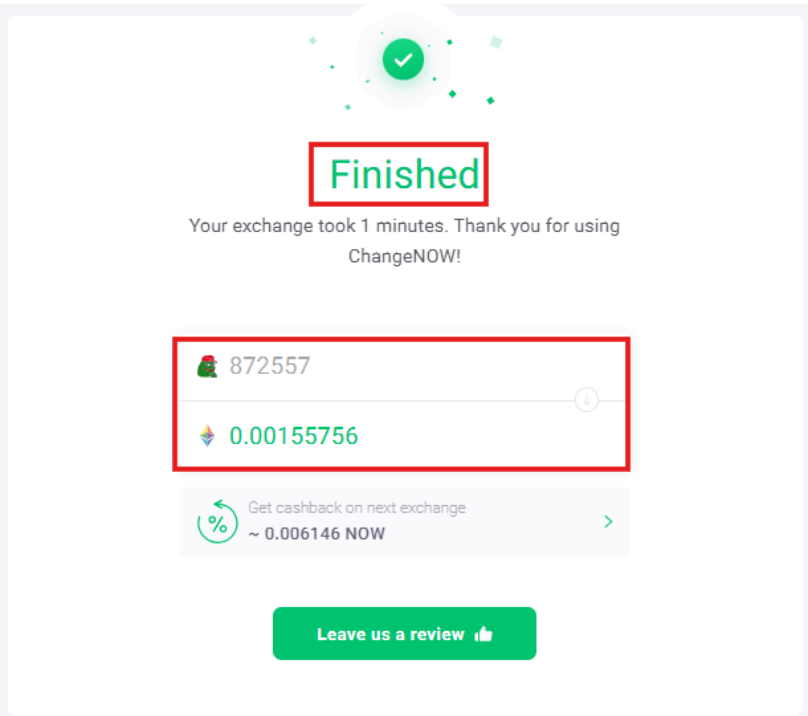
- **STEP 3) Analysis of Michael PC**

- Through analysis of Michael's PC, the following information was identified:

1. **Name of the cryptocurrency exchange service used:** ChangeNow
2. **Transaction creation time:** 2026-01-14T16:02:33
3. **Transaction ID:** 0x62ff5519fb2d8

Solution to Question 2

- STEP 4) Analysis of between Michael and Daniel's Transaction
 - It was confirmed that the transaction and exchange were conducted using **PEPE** and **ETH**.



Transaction details



Exchange order info

Completed time	14.01.2026, 16:17:42
Exchange Rate	1 PEPE = 0 ETH
Transaction ID	62ff5519f9b2d8

Input transaction info

Tx hash	0xc1e5ae5e60933ac52b0e0e79b833e0eb59c62be10b393ca8522e1ba291b3301b
Amount	872557 PEPE
Received time	14.01.2026, 16:16:32

Output transaction info

Tx hash	0x066eab41ee0d18fead2ba208d2f1b11657ea05492999ecf8b09af6a54d636c36
Your ETH Address	0xf2e5f287a02d0b10cb672bf6a99f237bb01959c4
Amount	0.00155756 ETH

Solution to Question 2

STEP 5) Transaction Tracing

1: from Dec ▾ → 0xF3d8F38c01Af8A7506fcf771BB04bAe3365aDC2c
2: to Dec ▾ → 0xAd3207e6268c2a94Cc697E7255d78D97d8a493f9

1: from Dec ▾ → 0xAd3207e6268c2a94Cc697E7255d78D97d8a493f9
2: to Dec ▾ → 0xdd3d72C53Ff982ff59853da71158bf1538b3Ceee

Convert PEPE to ETH

From: 0xEbA88149813BEc1cCcccFDb0daCEfaaa5DE94cB1 (ChangeNOW 16)
To: 0xf2E5F287A02d0B10cb672bf6A99f237bb01959C4

Daniel deposited funds into **Bithumb**.

The funds were transferred to a **one-based address**.

From the temporary account, the funds were deposited into the **official PEPE account**.

The PEPE was then converted to **ETH** and deposited into **ChangeNow**.

Finally, the funds were transferred from **ChangeNow** to **Michael**.

Solution to Question 2

■ STEP 5) Transaction Tracing

- Through transaction tracing, the following information was identified:

4. Michael's wallet address: 0xf2e5f287a02d0b10cb672bf6a99f237bb01959c4

5. Types of cryptocurrency used in the transaction: PEPE, ETH

6. Relevant dates and times of the transaction or exchange attempts

- **Transaction creation time:** 2026-01-14T16:02:33
- **Daniel's deposit time to Bithumb:** 2026-01-14T16:16:32
- **Time when ETH was received in Michael's wallet:** 2026-01-14T16:17:42

Solution to Question 2

- STEP 6) Analysis Daniel's Criminal Plan
 - Analysis of the PNG file containing the criminal plan, which Daniel attached via email, revealed that the file was **actually a PDF**.
 - Accordingly, the file extension was changed from **PNG to PDF**.

Re: Plan

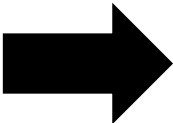
끝: Michael <michael.axisdefense@gmail.com>

plan-protected.png (533.82 KB)

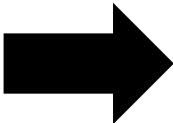
➔

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	25	50	44	46	2D	31	2E	36	0D	25	E2	E3	CF	D3	0D	0A	%PDF-1.6.ªªªªİÖ..
00000010	37	20	30	20	6F	62	6A	0A	3C	3C	2F	4C	69	6E	65	61	7 0 obj.<</Linea
00000020	72	69	7A	65	64	20	31	2F	4C	20	35	34	36	36	33	33	rized l/L 546633
00000030	2F	4F	20	31	30	2F	45	20	35	34	32	37	36	32	2F	4E	/O 10/E 542762/N
00000040	20	31	2F	54	20	35	34	36	33	33	32	2F	48	20	5B	20	l/T 546332/H [
00000050	35	37	39	20	32	32	34	5D	3E	3E	0A	65	6E	64	6F	62	579 224]>>.endob
00000060	6A	0A	20	20	20	20	20	20	20	20	20	20	20	20	20	20	j.

 plan-protected.png



 plan-protected.pdf



암호 입력

이 파일은 암호로 보호되어 있습니다. 파일을 열려면 암호를 입력하세요.

파일 열기

취소

Solution to Question 2

- Issue: PDF password could not be decrypted.
- Solve: Brute-force

```
import pikepdf
from multiprocessing import Pool, cpu_count
from tqdm import tqdm

def check_password(args):
    file_path, password = args
    try:
        with pikepdf.open(file_path, password=str(password)) as pdf:
            return str(password)
    except pikepdf.PasswordError:
        return None

def main():
    file_path = './plan-protected.pdf'
    start_num = 1402561
    end_num = 16777215

    num_cores = cpu_count()

    with Pool(num_cores) as pool:
        tasks = [(file_path, i) for i in range(start_num, end_num + 1)]
        for result in tqdm(pool.imap_unordered(check_password, tasks, chunksize=1000),
                           total=(end_num - start_num + 1)):
            if result:
                print(f"\n\n[성공] 비밀번호를 찾았습니다: {result}")
                pool.terminate()
                return

if __name__ == "__main__":
    main()
```

➡ [성공] 비밀번호를 찾았습니다: 12237685 ➡

Negotiation Agenda

Prepared By	Handled By	Scheduled Date
D.	M.	2026.01.15

Target

- Product development-related information

File Types

- Any available format

Transfer Method

- USB

Note

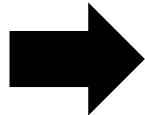
- Execute the operation after all employees have left the office
- Leave no unnecessary traces

Solution to Question 2

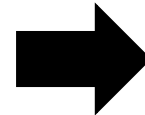
- Issue: PDF password could not be decrypted.
- Solve: Additional Information Provided

-최종 정리-
Q2. 마지막 문제.
A detailed description of Daniel's criminal plan 풀이 시,
본인이 생각하시는 아래 3가지 답안을 "길하민" 개인
톡으로 알려주시면 **추가 자료** 제공해드리겠습니다.

1. Michael 개인 지갑 주소
2. 파일형식
3. (Michael 지갑 주소로 만든)비밀번호



제가 생각하는 답안은 아래와 같습니다.
1. Michael 주소: 0xf2E5F287A02d0B10cb672bf6A99f237bb01959C4
2. 파일형식: PDF
3. 비밀번호: 1661380



Negotiation Agenda

Prepared By	Handled By	Scheduled Date
D.	M.	2026.01.15

Target

- Product development-related information

File Types

- Any available format

Transfer Method

- USB

Note

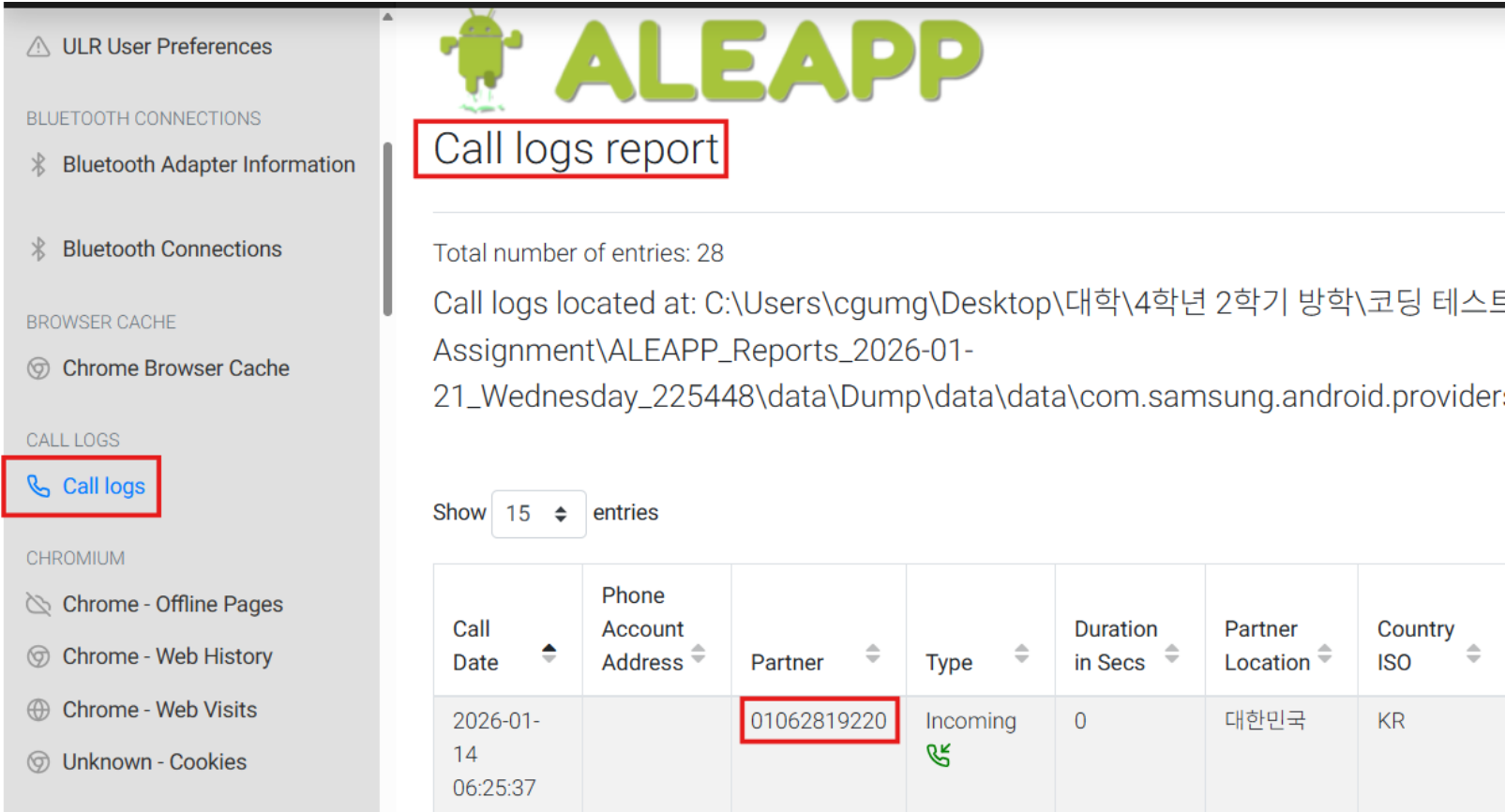
- Execute the operation after all employees have left the office
- Leave no unnecessary traces

Solution to Question 2

- **STEP 6) Analysis Daniel's Criminal Plan**
 - The following information can be found from the PDF:
 1. **Scheduled Date: 2026-01-15**
 2. **Target: Product development-related information**
 3. **File Types: Any available format**
 4. **Transfer Method: USB**
 5. **Note**
 - **Execute the operation after all employees have left the office**
 - **Leave no unnecessary traces**

Solution to Question 2

- STEP 7) Additional Artifacts
 - Additional artifact analysis revealed **traces of Michael and Daniel's phone calls.**



The screenshot shows the ALEAPP (Android Log Explorer and Parser) interface. On the left sidebar, the 'CALL LOGS' section is expanded, and 'Call logs' is selected. The main area displays the 'Call logs report' for a device. It indicates a total of 28 entries and provides the file path where the logs are located: C:\Users\cgumg\Desktop\대학\4학년 2학기 방학\코딩 테스트 Assignment\ALEAPP_Reports_2026-01-21_Wednesday_225448\data\Dump\data\data\com.samsung.android.provider:.

Below the text, there is a table showing the call log entries. The first entry is highlighted with a red box around the 'Partner' column value '01062819220'.

Call Date	Phone Account Address	Partner	Type	Duration in Secs	Partner Location	Country ISO
2026-01-14 06:25:37		01062819220	Incoming	0	대한민국	KR

Solution to Question 2

Q2	① Gmail, Telephony, MMS & SMS ② ChangeNow ③ 0x62ff5519fb2d8 ④ 0xf2e5f287a02d0b10cb672bf6a99f237bb01959c4 ⑤ PEPE, ETH (Get PEPE deposit and switch ETH from ChangeNow) ⑥ Transaction creation time: 2026-01-14T16:02:33 PEPE received time: 2026-01-14T16:16:32 ETH changed time: 2026-01-14T16:17:42 ⑦ Daniel's Criminal Plan - Scheduled Date: 2026.01.15 - Target: Product development-related information - File Types: Any available format - Transfer Method: USB - Note 1. Execute the operation after all employees have left the office 2. Leave no unnecessary traces
----	--

Questions

- Q3. Provide information about the AI server and where to get the information.

Solution to Question 3

Forensic Science International: Digital Investigation 54 (2025) 301987



Contents lists available at [ScienceDirect](#)

Forensic Science International: Digital Investigation

journal homepage: www.elsevier.com/locate/fsidi



DFRWS APAC 2025 - Selected Papers from the 5th Annual Digital Forensics Research Conference APAC

LangurTrace: Forensic analysis of local LLM applications

Sungjo Jeong, Sangjin Lee, Jungheum Park^{*}

School of Cybersecurity, Korea University, Seoul, South Korea

ARTICLE INFO

Keywords:

Digital forensics

Large language model (LLM)

Local LLM

Forensic artifact

Forensic tool development

ABSTRACT

A wide variety of applications have been developed to simplify the use of Large Language Models (LLMs), raising the importance of systematically analyzing their forensic artifacts. This study proposes a structured framework for LLM application environments, categorizing applications into backend runtime, client interface, and integrated platform components. Through experimental analysis of representative applications, we identify and classify artifacts such as chat records, uploaded files, generated files, and model setup histories. These artifacts provide valuable insight into user behavior and intent. For instance, LLM-generated files can serve as direct evidence in criminal investigations, particularly in cases involving the creation or distribution of illicit media, such as CSAM. The structured environment model further enables investigators to anticipate artifacts even in applications not directly analyzed. This study lays a foundational methodology for LLM application forensics, offering practical guidance for forensic investigations. To support practical adoption and reproducibility, we also release LangurTrace, an open-source tool that automates the collection and analysis of these artifacts.

jeongramon / LangurTrace

<> Code Issues Pull requests Actions Projects Security Insights

LangurTrace Public Watch 0

main 1 Branch 0 Tags

Go to file Add file Code

jeongramon Update README.md 3ed4648 · 6 months ago 5 Commits

dist	Add dist	6 months ago
resources	Initial Add	6 months ago
sample_dataset	Reupload sample dataset	6 months ago
src	Initial Add	6 months ago

Solution to Question 3

- STEP 1) Analysis of AI Server
 - Analysis of the AI server revealed download traces and artifacts related to **LM Studio**.

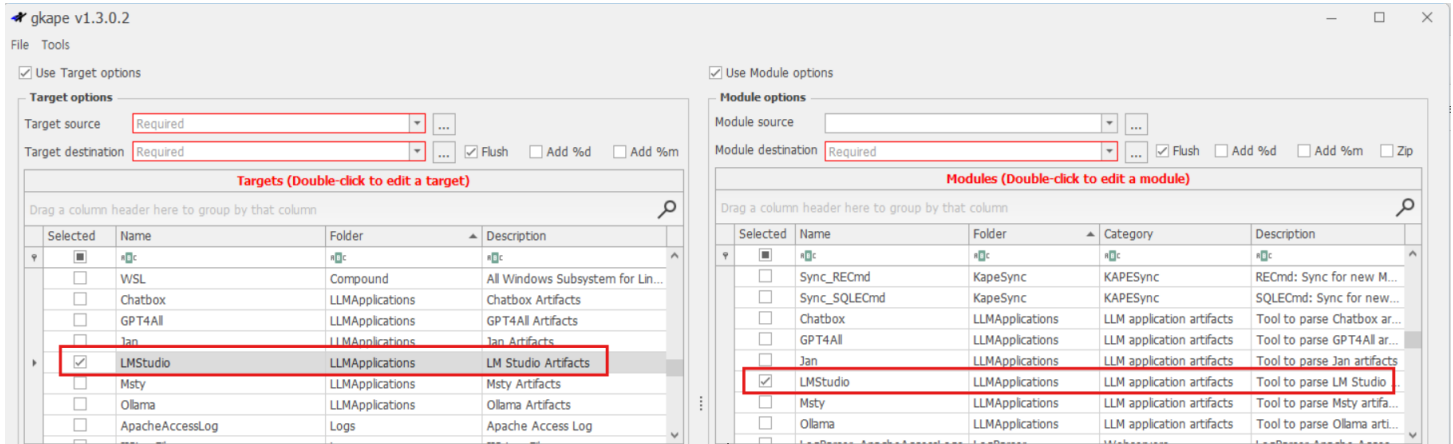
C:\Users\Administrator\Downloads\LM-Studio-0.3.37-1-x64.exe <https://lmstudio.ai/download/latest/win32/x64> 2026-01-14 13:57:46 KST



Listing						
/img_Server_PC.001/vol_vol6/Users/Administrator/.lmstudio						
Table Thumbnail Summary						
Name	S	C	O	Modified Time	Change Time	Access Time
.internal				2026-01-14 14:11:54 KST	2026-01-14 14:11:54 KST	2026-01-14 14:11:54 KST
[current folder]				2026-01-14 14:01:20 KST	2026-01-14 14:01:20 KST	2026-01-14 14:01:20 KST
[parent folder]				2026-01-15 18:37:36 KST	2026-01-15 18:37:36 KST	2026-01-15 18:37:36 KST
bin				2026-01-14 14:01:16 KST	2026-01-14 14:01:16 KST	2026-01-14 14:01:16 KST
config-presets				2026-01-14 14:01:15 KST	2026-01-14 14:01:15 KST	2026-01-14 14:01:15 KST
conversations				2026-01-14 14:23:19 KST	2026-01-14 14:23:19 KST	2026-01-14 14:23:19 KST
credentials				2026-01-14 14:01:15 KST	2026-01-14 14:01:15 KST	2026-01-14 14:01:15 KST
extensions				2026-01-14 14:01:15 KST	2026-01-14 14:01:15 KST	2026-01-14 14:01:32 KST
hub				2026-01-14 14:01:15 KST	2026-01-14 14:01:15 KST	2026-01-14 14:01:15 KST
mcp.json			1	2026-01-14 14:01:20 KST	2026-01-14 14:01:20 KST	2026-01-14 14:01:20 KST
models				2026-01-14 14:10:32 KST	2026-01-14 14:10:32 KST	2026-01-14 14:10:32 KST
server-logs				2026-01-14 14:01:45 KST	2026-01-14 14:01:45 KST	2026-01-14 14:01:45 KST
user-files				2026-01-14 14:16:06 KST	2026-01-14 14:16:06 KST	2026-01-14 14:16:06 KST
working-directories				2026-01-14 14:23:18 KST	2026-01-14 14:23:18 KST	2026-01-14 14:23:18 KST

Solution to Question 3

- **STEP 2) Using LangurTrace**
 - By using **LangurTrace**, artifacts related to **LM Studio** were extracted and analyzed.



Artifacts

📁 .internal	2026-01-23 오후 6:18	파일 폴더
📁 Conversations	2026-01-23 오후 6:18	파일 폴더
📁 models	2026-01-23 오후 6:18	파일 폴더
📁 user-files	2026-01-23 오후 6:18	파일 폴더

Analysis Results

📁 conversations	2026-01-23 오후 6:19	파일 폴더
📁 uploaded_files	2026-01-23 오후 6:19	파일 폴더
📄 model_setup_history.csv	2026-01-23 오후 6:19	Microsoft Excel 싼... 1KB
📄 models_index.csv	2026-01-23 오후 6:19	Microsoft Excel 싼... 1KB
📄 uploaded_files.csv	2026-01-23 오후 6:19	Microsoft Excel 싼... 1KB

Solution to Question 3

- STEP 3) Check the analysis results.
 - Conversations & Upload Files

Attached are the meeting minutes concerning our new product launch. Please write a report accordingly.

Step 1 - -

[1768367766660 - 229.docx](#)

Step 1 - -

MEETING MINUTES

Meeting Title	ITCS-X Development Proposal Review Meeting
Date & Time	January 10, 2026 (Friday), 14:00 - 16:30
Location	Main Conference Room, 4th Floor, Headquarters
Attendees	John Doe, Executive Managing Director (Business Division) Sarah Park, Director (Technology Research Institute) David Kim, Director (Strategic Planning Office) James Park, General Manager (Principal System Architect, Technical Development Division)
Minutes By	James Park

Solution to Question 3

- STEP 3) Check the analysis results.
 - Model Information

	A	B
1	Job ID	Model Name
2	modelDov	Goekdeniz-Guelmez/Josiefied-Qwen2.5-3B-Instruct-abliterated-v1-gguf/josiefied-qwen2.5-3b-instruct-abliterated-v1-q4_k_s.gguf

Solution to Question 3

■ STEP 3) Analysis Results Check

- As a result of the AI server analysis, the following information was found:

1. **AI Platform used:** LM Studio (from Web History, Web Download)

2. **Found artifacts**

- **Downloaded models:** Qwen 2.5-based custom models (from C:/Users/Administrator/.lmstudio/internal/download-jobs-info.json)
- **Model Information:** (from C:/Users/Administrator/models/)
- **Conversation history:** We can view conversation history that asks LLM to write a report based on minutes James have written (from C:/Users/Administrator/.lmstudio/conversations/)
- **Upload file:** ITCS-X_Meeting_Minutes.docx (from C:/Users/Administrator/.lmstudio/user-file/)

3. **James's Role: General Manager (Principal System Architect, Technical Development Division)**

Solution to Question 3

Q3	Information about an AI server 1. AI Platform used: LM Studio (from Web History, Web Download) 2. Found artifacts - Downloaded models: Qwen 2.5-based custom models (from C:/Users/Administrator/.lmstudio/.internal/download-jobs-info.json) - Model Information: (from C:/Users/Administrator/models/) - Conversation history: We can view conversation history that asks LLM to write a report based on minutes James have written (from C:/Users/Administrator/.lmstudio/conversations/) - Upload file: ITCS-X Meeting Minutes.docx (from C:/Users/Administrator/.lmstudio/user-file/)
----	--

Questions

- Q4. Describe how Michael leaked data from the AI server.

Solution to Question 4

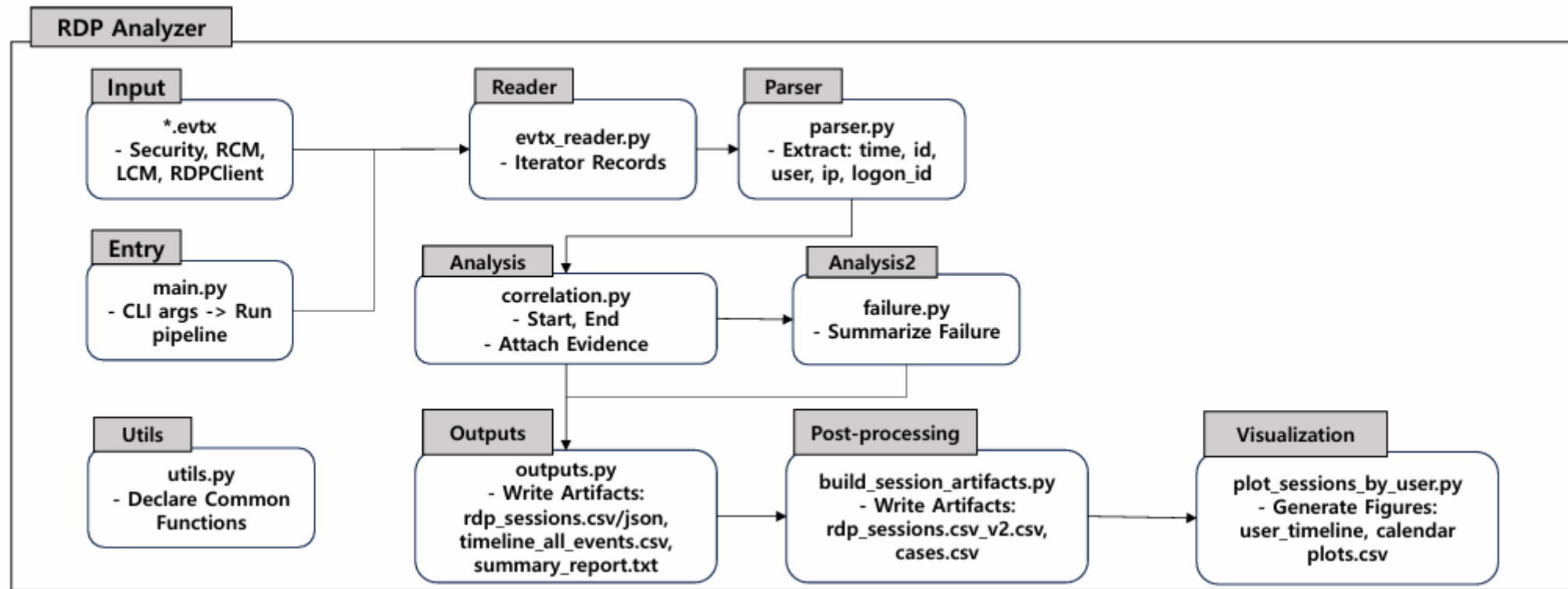
■ STEP 1) Review of Email Contents and Daniel's Criminal Plan

- Michael stated that he does **not have sufficient privileges on his own account** and therefore must access the system **using another account**.
- He mentioned that the AI server likely **contains conversation logs and materials related to new technologies**.
- Daniel stated that the data would be transferred via a **USB device**.
- He also indicated that the plan would be executed after all **employees had left**.
- Additionally, Michael was aware of **James's PC password**.
 - *"Therefore, it can be inferred that **Michael used James's PC to access the AI server**."*
 - *"It is reasonable to suspect that **Remote Desktop Protocol (RDP)** was used via James's PC."*
 - *"The access likely occurred after working hours, presumably **after 18:00**."*

Solution to Question 4

■ STEP 2) Analysis of the AI Server

- The **RDP Analyzer Program**, which was used in Week 1 Seminar Assignment Problem 1, was used.



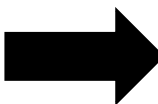
Solution to Question 4

■ STEP 2) Analysis of the AI Server

- The AI server **event logs** were extracted and analyzed.
- As a result, it was confirmed that a user from the IP address '**192.168.220.136**' accessed the server.

```
Parsing Security: 818it [00:06, 122.43it/s]
Parsing LSM: 48it [00:00, 211.89it/s]
Parsing RCM: 23it [00:00, 217.86it/s]

=== DONE ===
[+] Sessions CSV: output/rdp_sessions.csv
[+] Sessions JSON: output/rdp_sessions.json
[+] Timeline CSV: output/timeline_all_events.csv
[+] Summary Report: output/summary_report.txt
```



```
=== RDP Analysis Summary Report ===

Total RDP Success Sessions: 1
Unique Usernames: 1
Unique Client IPs: 1

[Top Client IPs from Successful Sessions]
- 192.168.220.136: 1 sessions
```

Solution to Question 4

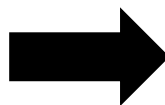
■ STEP 2-1) Analysis of James's PC

- Execution traces of the **mstsc.exe** program were identified on James's PC.
- Accordingly, the event logs of James's PC were analyzed using the **RDP Analyzer**, which revealed session creation traces associated with the IP address '**192.168.220.137**'.

 Documents.Ink			C:\Users\James Park\Documents
 mstsc.exe.Ink			C:\Windows\System32\mstsc.exe
 No preferred path found.Ink			No preferred path found

```
Parsing Security: 21522it [03:10, 113.12it/s]
Parsing LSM: 150it [00:00, 251.93it/s]
Parsing RCM: 30it [00:00, 189.56it/s]

=== DONE ===
[+] Sessions CSV: output/rdp_sessions.csv
[+] Sessions JSON: output/rdp_sessions.json
[+] Timeline CSV: output/timeline_all_events.csv
[+] Summary Report: output/summary_report.txt
```



```
=== RDP Analysis Summary Report ===

Total RDP Success Sessions: 1
Unique Usernames: 1
Unique Client IPs: 1

[Top Client IPs from Successful Sessions]
- 192.168.220.137: 1 sessions
```

Solution to Question 4

■ STEP 2-2) Michael's PC 분석

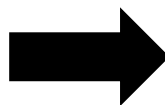
- Execution traces of the **mstsc.exe** program were also identified on Michael's PC.
- Accordingly, the event logs of Michael's PC were analyzed using the **RDP Analyzer**, but **no evidence of an RDP connection** was found.

 Documents.lnk			C:\Users\James Park\Documents
 mstsc.exe.lnk			C:\Windows\System32\mstsc.exe
 No preferred path found.lnk			No preferred path found

```
Parsing Security: 21418it [02:29, 143.48it/s]
Parsing LSM: 130it [00:00, 179.88it/s]
Parsing RCM: 12it [00:00, 239.35it/s]
```

```
=== DONE ===
```

```
[+] Sessions CSV: output/rdp_sessions.csv
[+] Sessions JSON: output/rdp_sessions.json
[+] Timeline CSV: output/timeline_all_events.csv
[+] Summary Report: output/summary_report.txt
```



```
=== RDP Analysis Summary Report ===
```

```
Total RDP Success Sessions: 0
Unique Usernames: 0
Unique Client IPs: 0
```

```
[Top Client IPs from Successful Sessions]
```

```
[Users from Successful Sessions]
```

Solution to Question 4

■ Findings from RDP Analyzer Analysis

- Michael's PC did not receive any RDP connections; therefore, it functioned as a **client**.
- James's PC received an RDP connection and functioned as a **server**.
- The AI server also received an RDP connection and functioned as a **server**.

→ *"However, since Michael did not have authorization to access the AI server directly, it is inferred that he **first established an RDP connection to James's PC** and then **accessed the AI server from there**."*

→ *"In other words, the access path followed the structure: **Michael's PC → James's PC → AI Server**"*

Solution to Question 4

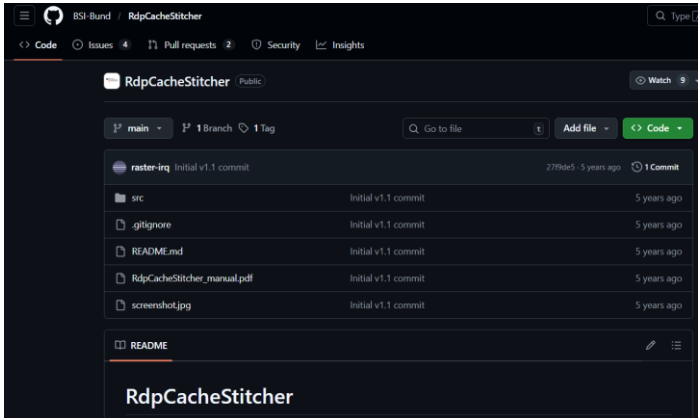
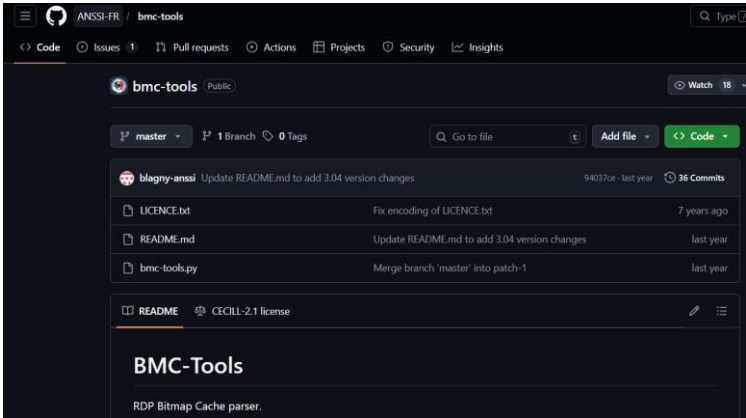
- STEP 3) Additional Artifact Analysis of Michael's and James's PCs
 - During RDP execution, **RDP cache** files are stored in '**C:\Users\Michael\AppData\Local\Microsoft\Terminal Server Client\Servers**', and these artifacts were analyzed.
 - Analysis tools:** bmc-tools, RdpCacheStitcher

/img_Michael_PC.001/vol_vol6/Users/Michael Lee/AppData/Local/Microsoft/Terminal Server Client/Cache

Name	S	C	O	Modified Time	Change Time	Access Time
Cache0000.bin			0	2026-01-15 18:36:25 KST	2026-01-15 18:36:25 KST	2026-01-15 18:36:47 KST
Cache0001.bin			0	2026-01-15 18:39:35 KST	2026-01-15 18:39:35 KST	2026-01-15 18:39:35 KST
[current folder]				2026-01-15 18:39:35 KST	2026-01-15 18:39:35 KST	2026-01-15 18:39:35 KST
[parent folder]				2026-01-15 18:36:07 KST	2026-01-15 18:36:07 KST	2026-01-15 18:36:39 KST
bcache24.bmc				2026-01-15 18:36:19 KST	2026-01-15 18:36:19 KST	2026-01-15 18:36:19 KST

/img_James_PC.001/vol_vol6/Users/James Park/AppData/Local/Microsoft/Terminal Server Client/Cache

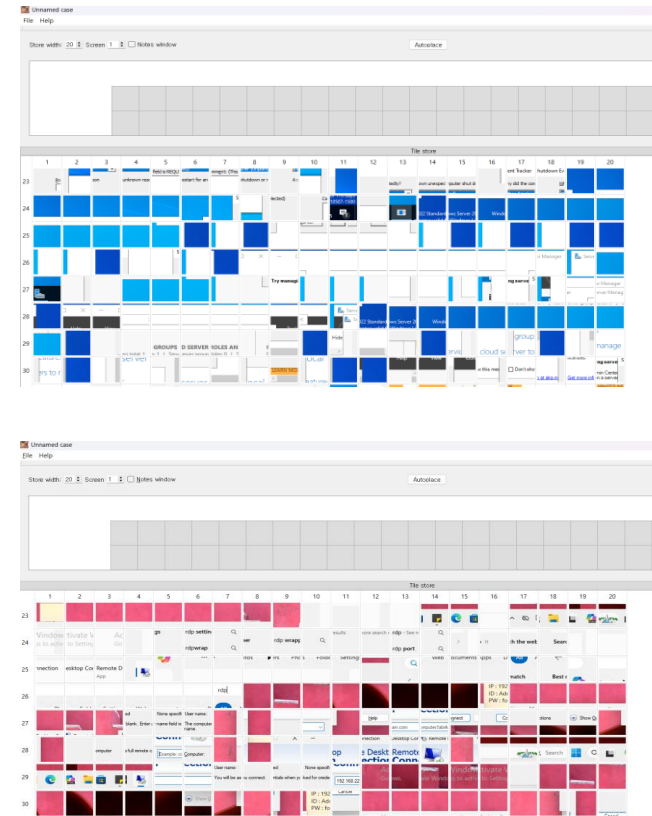
Name	S	C	O	Modified Time	Change Time	Access Time
Cache0000.bin			0	2026-01-15 18:39:30 KST	2026-01-15 18:39:30 KST	2026-01-15 18:39:30 KST
[current folder]				2026-01-15 18:39:30 KST	2026-01-15 18:39:30 KST	2026-01-15 18:39:30 KST
[parent folder]				2026-01-15 18:37:12 KST	2026-01-15 18:37:12 KST	2026-01-15 18:37:12 KST
bcache24.bmc				2026-01-15 18:37:35 KST	2026-01-15 18:37:35 KST	2026-01-15 18:37:35 KST



Solution to Question 4

- **STEP 3) Additional Artifact Analysis of Michael's and James's PCs**
 - First, **bmc-tools** was used to convert **.bin** files to **.bmp** format, and the resulting **BMP** files were then combined using **RdpCacheStitcher**.

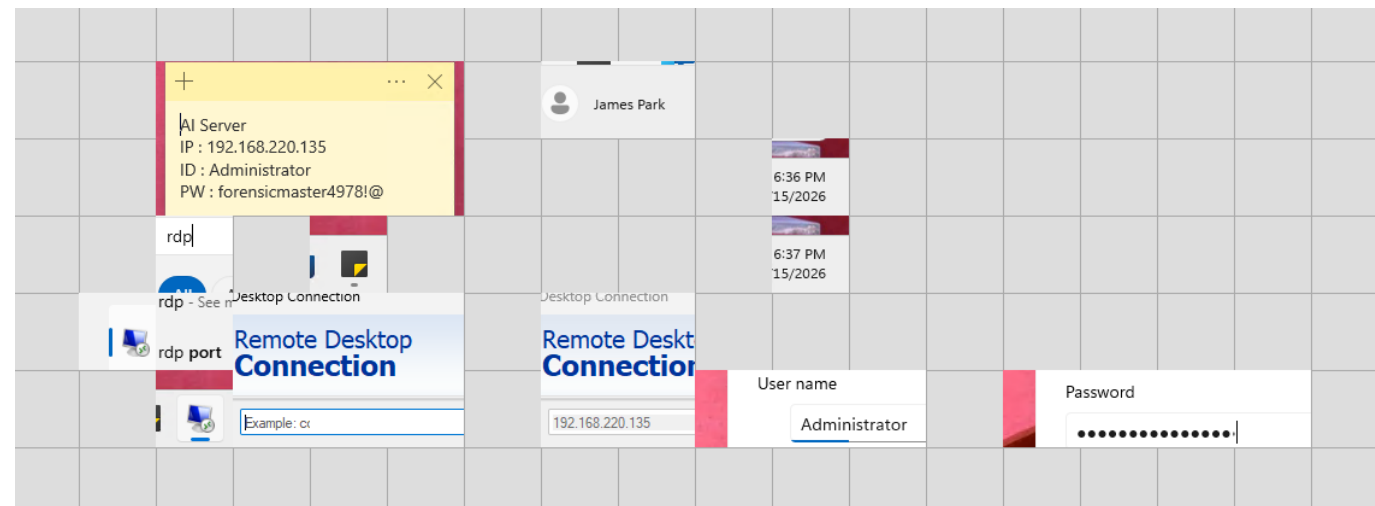
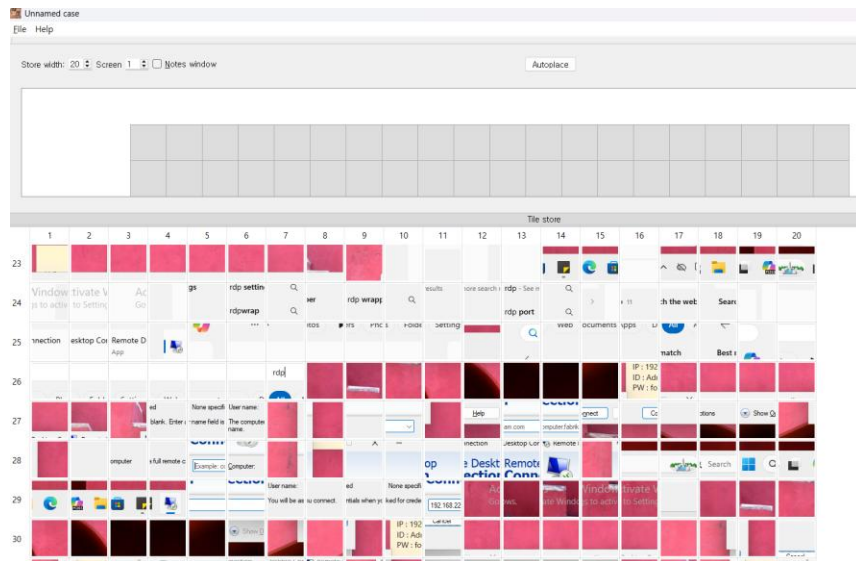
```
0000.bmp
0001.bmp
0002.bmp
0003.bmp
0004.bmp
0005.bmp
0006.bmp
0007.bmp
0008.bmp
0009.bmp
0010.bmp
0011.bmp
0012.bmp
```



Solution to Question 4

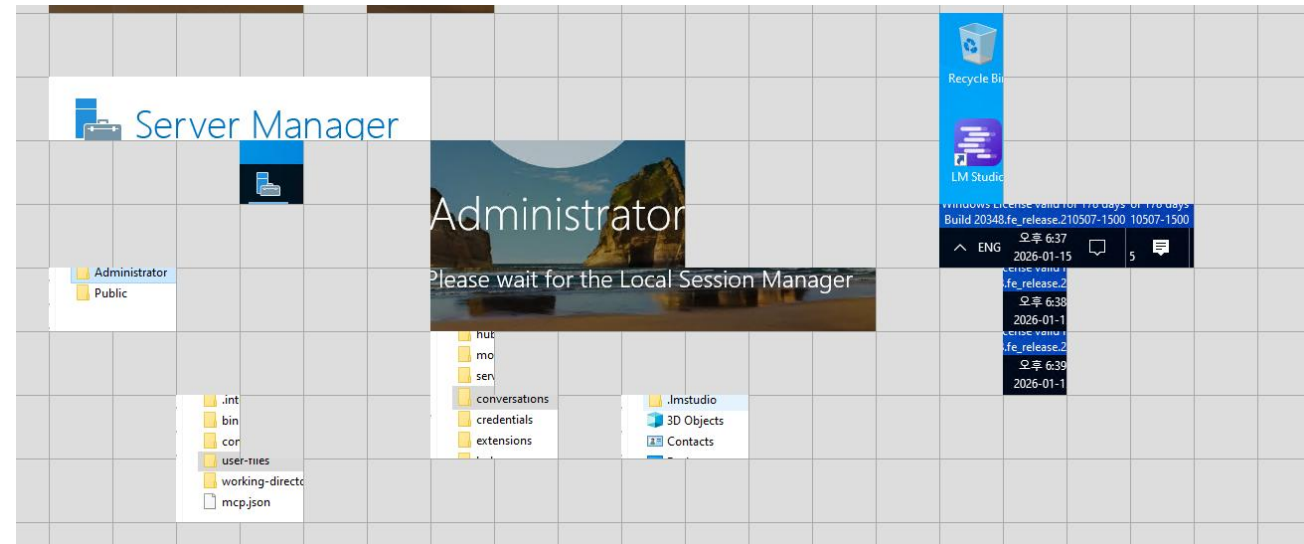
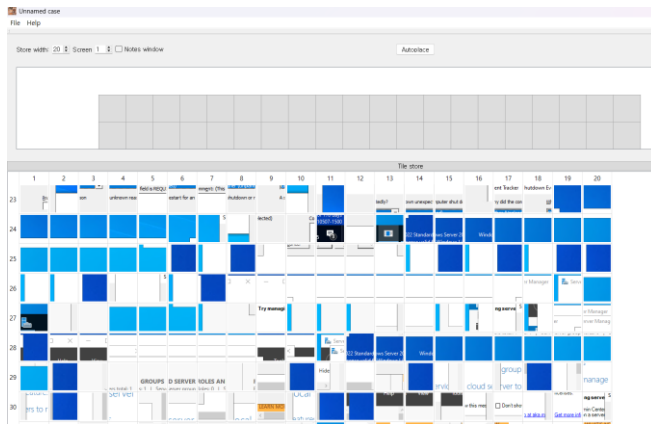
■ STEP 3-1) Reconstruction of RDP Cache on Michael's PC

- As a result of the reconstruction, it was confirmed that Michael's PC had previously established an RDP connection to James's PC, and the AI server's IP address and password were visible on the screen of that PC.
- This provides evidence supporting the assumption that Michael reconnected to James's PC via RDP and subsequently accessed the AI server.



Solution to Question 4

- **STEP 3-2) Reconstruction of RDP Cache on James's PC**
 - On James's PC, it was determined that an **Administrator** account was used to access the AI server, and AI-related information was accessed through **Server Manager**.



Solution to Question 4

STEP 4) Additional Analysis of James's PC

- The presence of **RDPCLIP.exe** was identified in the **Prefetch** artifacts on James's PC.
- This indicates that Michael transferred AI-related data from the AI server to his own PC via **drag-and-drop** during the RDP session.

* Michael accessed the **conversations, user-files** directory and retrieved conversation logs and uploaded files.

/img_James_PC.001/vol_vol6 Windows/Prefetch

Table Thumbnail Summary						
^ Name	S	C	O	Modified Time	Change Time	Access Time
ONEDRIVESETUP.EXE-2F4C4906.pf			0	2026-01-14 15:06:18 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
ONEDRIVESETUP.EXE-339C4628.pf			0	2026-01-14 15:07:18 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
ONEDRIVESETUP.EXE-B9CEA7F7.pf			0	2026-01-15 13:42:24 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
OOBENETWORKCONNECTIONFLOW.EXE-ED695E			0	2026-01-14 15:02:06 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
OPENCONSOLE.EXE-DDB337D4.pf			0	2026-01-15 13:45:13 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
OPENWITH.EXE-8B50D58B.pf			0	2026-01-14 17:19:06 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
POWERSHELL.EXE-CA1AE517.pf			0	2026-01-15 14:25:07 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
PfPre_f31617e9.mkd			0	2026-01-13 18:30:35 KST	2026-01-16 14:19:29 KST	2026-01-15 16:18:23 KST
PfPre_f316ddc6.mkd			0	2026-01-14 15:03:34 KST	2026-01-16 14:19:29 KST	2026-01-16 14:19:29 KST
RDPCLIP.EXE-7D8DB38B.pf			0	2026-01-15 18:36:49 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST
REG.EXE-A93A1343.pf			0	2026-01-14 14:57:55 KST	2026-01-16 14:19:29 KST	2026-01-15 18:40:21 KST

Solution to Question 4

- **STEP 5) Additional Analysis of Michael's PC**
 - Analysis of the **Recent Documents** on Michael's PC showed a correlation between the '**E: drive**' and **AI-related files**.
 - This suggests that Michael exfiltrated AI data that had been transferred from the AI server to his PC via **drag-and-drop**, using a **USB device**.

Recent Documents						
Table Thumbnail Summary						
Source Name	S	C	O	Path	▼ Date Accessed	
 plan-protected.Ink				C:\Users\Michael Lee\Downloads\plan-protected....	2026-01-14 16:36:49 KST	
 plan-protected.Ink				C:\Users\Michael Lee\Downloads\plan-protected....	2026-01-14 16:36:49 KST	
 plan-protected.Ink				C:\Users\Michael Lee\Downloads\plan-protected....	2026-01-14 16:36:49 KST	
 No preferred path found.Ink				No preferred path found	0000-00-00 00:00:00	
 user-files.Ink				E:\user-files	0000-00-00 00:00:00	
 conversations.Ink				E:\conversations	0000-00-00 00:00:00	

Solution to Question 4

Q4

STEP 1. Michael connects **RDP** from **his PC to James PC**

STEP 2. AI Server IP & Password found on James PC

STEP 3. RDP connection from **James PC to AI server**

STEP 4. Get conversation logs and new product or technology related records from LM Studio from the **AI server to Drag & Drop to Michael PC**

STEP 5. Exporting from Michael PC to the outside via **USB**

Questions

- **Q5. Create a table of who James contacted via Messenger and what conversations they had.**

Solution to Question 5

Forensic Science International: Digital Investigation 54 (2025) 301941

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journal homepage: www.elsevier.com/locate/fsidi



Decrypting messages: Extracting digital evidence from signal desktop for windows

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ARTICLE INFO

ABSTRACT

Keywords:
Digital forensics
Signal
Windows
Encryption
Electron
Instant messaging

With growing concerns over the security and privacy of personal conversations, end-to-end encrypted instant messaging applications have become a key focus of forensic research. This study presents a detailed methodology along with an automated Python script for decrypting and analyzing forensic artifacts from Signal Desktop for Windows. The methodology is divided into two phases: i) decryption of locally stored data and ii) analysis and documentation of forensic artifacts. To ensure data integrity, the proposed approach enables retrieval without launching Signal Desktop, preventing potential alterations. Additionally, a reporting module organizes extracted data for forensic investigators, enhancing usability. Our approach is effective in extracting and analyzing encrypted Signal artifacts, providing a reliable method for forensic investigations.

labcif / SignalForensics

<> Code Issues 1 Pull requests Actions Projects Security Insights

SignalForensics Public

main 2 Branches 0 Tags

Go to file Add file

gonssalu feat: include timestamp hash cdc9a85 · 4 months ago

images chore: README file

modules feat: include timestamp hash

.gitignore feat: process message status

.prettierrc feat: base syntax

Solution to Question 5

- STEP 1) James's PC Analysis
 - Artifacts related to **Signal Desktop** were identified.

/img_James_PC.001/vol_vol6

Users/James Park/AppData/Roaming/Signal

Table	Thumbnail	Summary				
△ Name	S	C	O	Modified Time	Change Time	Access Time
 Cache				2026-01-15 13:47:09 KST	2026-01-15 13:47:09 KST	2026-01-15 13:47:09 KST
 Code Cache				2026-01-15 13:47:09 KST	2026-01-15 13:47:09 KST	2026-01-15 13:47:09 KST
 Crashpad				2026-01-15 13:47:09 KST	2026-01-15 13:47:09 KST	2026-01-15 13:47:09 KST
 DIPS			0	2026-01-15 13:48:06 KST	2026-01-15 13:48:06 KST	2026-01-16 14:20:01 KST
 DIPS-wal			0	2026-01-15 17:47:13 KST	2026-01-15 17:47:13 KST	2026-01-15 17:47:13 KST
 DawnGraphiteCache				2026-01-15 13:47:12 KST	2026-01-15 13:47:12 KST	2026-01-16 14:20:00 KST
 DawnWebGPUCache				2026-01-15 13:47:12 KST	2026-01-15 13:47:12 KST	2026-01-16 14:20:00 KST
 GPUCache				2026-01-15 13:47:12 KST	2026-01-15 13:47:12 KST	2026-01-16 14:20:00 KST

Solution to Question 5

- STEP 2) Using SignalForensics

```
SignalDecryptor v3.1.2 by @gonssalu
-----==<[ CONFIG ]>=====
Mode: aux
Environment: windows
Signal Directory: ..\Signal
Output Directory: ..\Signal_output
-----

[i] SQLCipher Key: c556a89fab4315020aa55dc0d442bf100696f8c496a3954a157138be09191052
[i] Opening SQLCipher database
[i] Exported the unencrypted database
[i] Exporting attachments...
[i] Exported 0 attachments
[i] Writing reports...
[i] No call logs were found in the database
```



	9ef63220-6788-459c-bbe5-766459544...	2026-01-25 오후 4:00	파일 폴더	
	019bbffb-2545-7458-8943-04d6e5093...	2026-01-25 오후 4:00	파일 폴더	
	contacts.csv	2026-01-25 오후 4:00	Microsoft Excel 쉼...	1KB
	conversations.csv	2026-01-25 오후 4:00	Microsoft Excel 쉼...	1KB
	files_hashes.csv	2026-01-25 오후 4:00	Microsoft Excel 쉼...	1KB
	full_report.html	2026-01-25 오후 4:00	Chrome HTML Do...	27KB

Questions

- Q5. Create a table of who James contacted via Messenger and what conversations they had.

Time	Sent by	Content	Note
26.01.13T16:41:55	Daniel → James	Regarding what I mentioned last time are we ready to move forward with execution?	...
26.01.13T16:44:39	James → Daniel	Moving directly carries significant risk. However, there's an assistant manager named Michael. He seems to be in a tight spot lately . It would be better to use him as a shield (or decoy) .	
26.01.13T16:45:59	Daniel → James	Sounds good.	
26.01.13T16:46:11	Daniel → James	If you can arrange a meeting, I'll try to recruit him .	
26.01.13T16:46:18	Daniel → James	Please pass along my business card to Michael	
26.01.13T16:47:07	Daniel → James	NaN	Attachments (Daniel's Business Card)
26.01.13T16:47:19	James → Daniel	Understood	
26.01.13T16:47:27	James → Daniel	I'll send you Michael's contact information and basic details now.	

Freshmen Mock Case Scenario (DFRC #51)

Questions

Time	Sent by	Content	Note
26.01.13T16:48:51	James → Daniel	Phone number: 010-5529-8452 Email: michael.axisdefense@gmail.com Position: Assistant Manager	...
26.01.13T16:49:19	James → Daniel	If the plan succeeds, my position will be secured , right?	
26.01.13T16:49:49	Daniel → James	Yes. I'll arrange for an executive director position for you.	
26.01.16T14:21:55	James → Daniel	We have a problem. What we were worried about has happened.	
26.01.16T14:22:20	James → Daniel	Michael was caught on-site while taking out the materials, and the USB has been secured as well.	
26.01.16T14:23:22	Daniel → James	The situation has taken a turn for the worst...	
26.01.16T14:23:35	Daniel → James	However, you still have access to that server's data, don't you?	
26.01.16T14:23:56	Daniel → James	If you can secure and deliver that data even now , I'll promise you compensation and an executive position that's incomparable to what we previously discussed . There's no reason to stop here .	
26.01.16T14:24:51	NaN	Understood. I'll write down the necessary information and deliver it to you.	Presumably James → Daniel
26.01.16T14:25:02	NaN	I'll contact you again once it's ready.	Presumably James → Daniel

Solution to Question 5

■ STEP 3) Artifact Analysis Results

- Based on the **Signal Desktop artifacts** from James's PC, the following information was identified:
 1. **James and Daniel had planned the crime** in advance.
 2. From one perspective, Michael can be considered a **victim of the criminal plan** devised by James and Daniel.
 3. Michael was ultimately **apprehended**.
 4. However, Daniel **did not stop the plan** and promised James a position as an executive director, along with compensation far exceeding anything previously offered.
 5. James **accepted the offer** and stated that he would make contact once preparations were complete.

Questions

- **Q6. Summarize the case based on all evidence collected so far.**
 - Submit: ① **Character details**, ② Full incident timeline, ③ Visual summary of the case
 - Follow the templates as closely as possible and provide all necessary details (multiple pages allowed)



Name : Michael Lee

Company : Axis Defense

Role : Assistance Manager

Artifacts :

- MMS/SMS Records
- Email Records
- James's PC Password
- Web Browser Artifacts
- RDP Artifacts (Event Log, Cache)
- Recent Documents
- USB Attachments Records



Name : James Park

Company : Axis Defense

Role : General Manager,
(Principal System
Architect, Technical
Development Division)

Artifacts :

- Signal Desktop Artifacts
- RDP Artifacts (Event Log, Cache)
- Prefetch Artifacts
- AI Server Artifacts



Name : Daniel

Company : NORTHRIDGE
Defense

Role : Director of Strategic
Programs, (Defense
System Division)

Artifacts :

- SMS/MMS Records
- Email Artifacts
- Criminal Plan

- Submit: ① Character details, ② **Full incident timeline**, ③ Visual summary of the case
 - Follow the templates as closely as possible and provide all necessary details (multiple pages allowed)

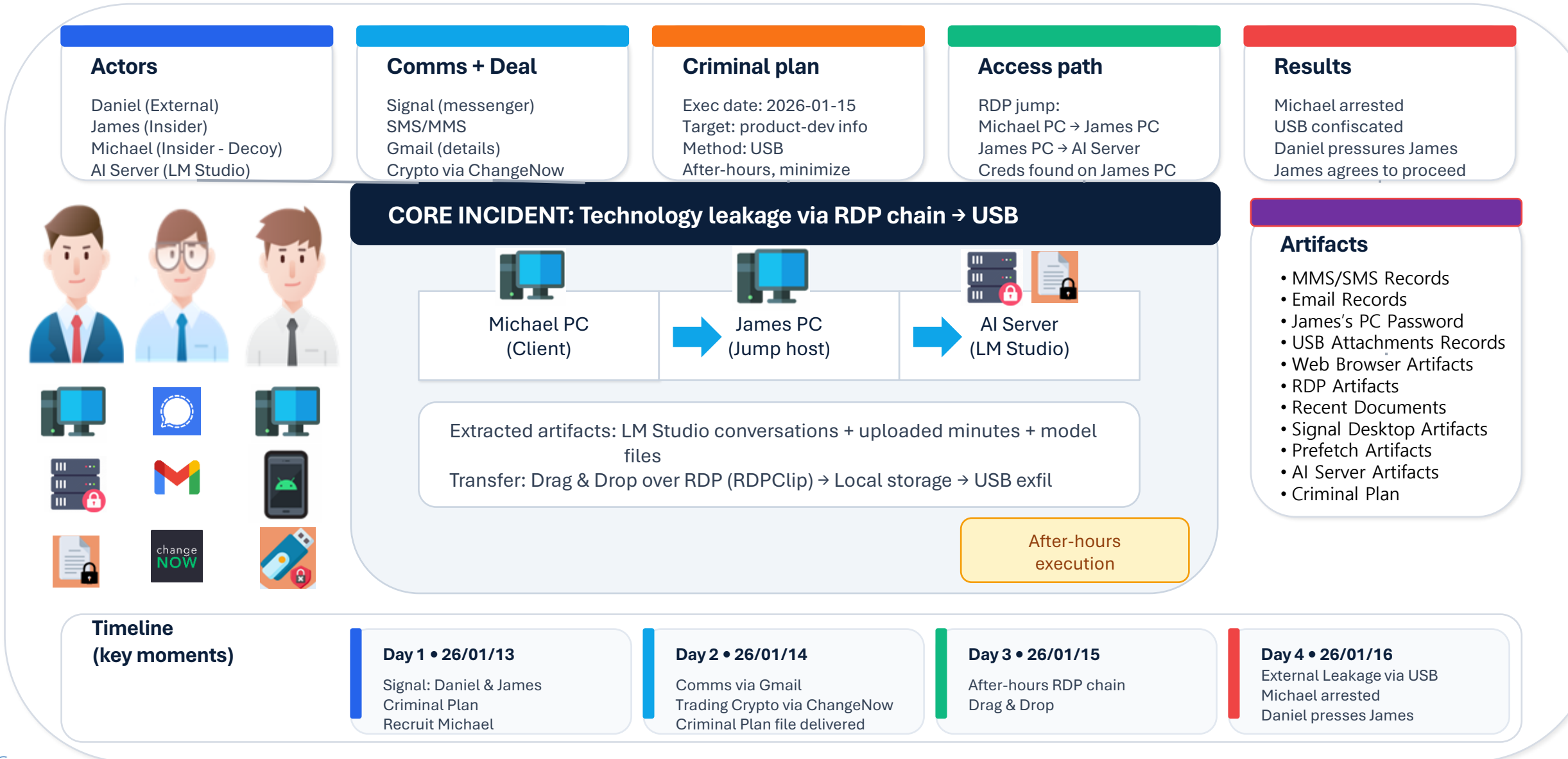


Questions

- **Q6. Summarize the case based on all evidence collected so far.**
 - Submit: ① Character details, ② Full incident timeline, ③ **Visual summary of the case**
 - Follow the templates as closely as possible and provide all necessary details (multiple pages allowed)

[③ Visual summary of the case] → **Diagram it freely** 😊

Freshmen Mock Case Scenario (DFRC #51)



Conclusion

Conclusion

- This assignment involved analyzing an incident response case and constructing a timeline of events.
- The incident could be reconstructed by examining artifacts discovered in the provided images and dump files.
- Although this case did not heavily involve anti-forensic techniques, many aspects could be analyzed in a relatively straightforward manner.
- It was considered that if anti-forensic techniques had been applied, the case analysis would have been significantly more challenging.
- Personally, this assignment was a meaningful challenge that allowed me to apply the forensic techniques I had previously learned.

THANK YOU
for Listening!



Digital Forensic Research Center

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디 지 털 포 렌 식 연 구 센 터

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Questions?

